



Efficacy and safety of once-weekly subcutaneous zenagamtide, a novel unimolecular GLP-1 and amylin receptor agonist, in type 2 diabetes: a multicentre, randomised, parallel, double-blind, placebo-controlled, dose-finding, phase 2 trial

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Summary

Background Zenagamtide (formerly amycletin) is a unimolecular agonist of GLP-1, amylin, and calcitonin receptors. We aimed to investigate the dose–response relationship with respect to the efficacy and safety of once-weekly subcutaneous zenagamtide compared with placebo in adults with type 2 diabetes.

Methods This 36-week, randomised, parallel, double-blind, placebo-controlled, dose-finding, phase 2 trial was conducted at 83 hospital and clinic sites across 11 countries. The trial consisted of a 3-week screening period, a 36-week intervention period, and a 4-week follow-up period. Adults (aged 18–75 years) with type 2 diabetes (glycated haemoglobin [HbA_{1c}] 7·0–10·0% [53–86 mmol/mol]), treated with stable doses of metformin with or without a SGLT2 inhibitor, and with a BMI of 23·0 to <50·0 kg/m² were randomly assigned (6:1) to once-weekly subcutaneous zenagamtide or placebo using a randomisation and trial supplies management system, before being further randomly assigned to one of six dose groups (1:1:1:1:1:1; 0·4, 1·5, 5, 10, 20, or 40 mg). Participants assigned to placebo were similarly allocated to matched placebo regimens corresponding to each dosing schedule. Randomisation was stratified by HbA_{1c} (<8·5% or ≥8·5%) at screening and country of residence (Japan or other countries). The starting dose of subcutaneous zenagamtide was 0·2 mg, with dose escalations every 4 weeks until the maintenance dose was reached (0·4–40 mg). Participants and staff were masked within each dose level to placebo or zenagamtide. The primary outcome was change in HbA_{1c} from baseline to week 36. The primary outcome was assessed in all randomly assigned participants using the efficacy estimand based on on-treatment data without rescue medication. Safety outcomes were assessed in all randomly assigned participants exposed to treatment. The trial is registered with ClinicalTrials.gov, NCT06542874, and is completed.

Findings Between Aug 7 and Dec 27, 2024, 262 (29%) of 915 screened individuals were randomly assigned to subcutaneous zenagamtide (n=225) or placebo (n=37). Participants were further randomly assigned (1:1:1:1:1:1) to one of six doses (38 participants to 0·4 mg, 36 participants to 1·5 mg, 37 participants to 5 mg, 38 participants to 10 mg, 38 participants to 20 mg, and 38 participants to 40 mg) or placebo (37 participants). 261 participants were exposed to treatment. At week 36, estimated change in HbA_{1c} (mean across all groups at baseline 7·8% [SD 0·8]) ranged from –0·9% with zenagamtide 0·4 mg (estimated treatment difference vs placebo: –0·77% [95% CI –1·26 to –0·28]; p=0·0021) to –1·7% with zenagamtide 40 mg (–1·56% [–2·05 to –1·07]; p<0·0001). Most adverse events were gastrointestinal and mild to moderate in severity. 21 (8%) of 261 participants reported serious adverse events (four with zenagamtide 0·4 mg, three with 1·5 mg, two with 5 mg, five with 10 mg, three with 20 mg, one with 40 mg, and three with placebo). No deaths occurred.

Interpretation In people with type 2 diabetes, once-weekly subcutaneous zenagamtide 0·4–40 mg demonstrated clinically meaningful and significant improvements in change in HbA_{1c} from baseline to week 36 compared with placebo, with a safety and tolerability profile consistent with that of other GLP-1-based and amylin-based therapies.

Funding Novo Nordisk.

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Introduction

International treatment guidelines emphasise the importance of weight management as part of a holistic

approach to type 2 diabetes management because weight loss has been linked to improved glycaemic control and better longitudinal cardiometabolic outcomes.^{1–3}

Lancet 2026; 408: 621–35

Published Online
July 30, 2026
[https://doi.org/10.1016/S0140-6736\(26\)01248-1](https://doi.org/10.1016/S0140-6736(26)01248-1)
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Research in context

Evidence before this study

GLP-1 and amylin receptor agonists exert complementary physiological effects that support improved glycaemic control, bodyweight reduction, and other cardiometabolic benefits through distinct but synergistic mechanisms. Combining these pathways has therefore emerged as a promising strategy to further benefit metabolic outcomes in individuals with overweight or obesity, with or without type 2 diabetes. We conducted a literature search in PubMed for studies published in any language between Feb 12, 2016, and Feb 13, 2026, using search terms including “glucagon-like peptide AND amylin AND type 2 diabetes” and “glucagon-like peptide-1 AND amylin AND obesity”. Identified publications included phase 3 clinical trials evaluating combined GLP-1 and amylin receptor agonism, such as the cagrilintide and semaglutide combination therapy CagriSema, which showed clinically meaningful reductions in bodyweight and improvements in glycaemic outcomes compared with placebo in individuals with overweight or obesity, including those with type 2 diabetes.

Zenagamtide (previously known as amycretin), a unimolecular peptide agonist of GLP-1, amylin, and calcitonin receptors, has been evaluated in early-phase clinical trials in individuals with overweight or obesity without type 2 diabetes. In a phase 1b/2a study, subcutaneous zenagamtide produced significant dose-dependent reductions in bodyweight compared with placebo, with a safety and tolerability profile consistent with that of GLP-1 and amylin receptor agonists. These findings provided early clinical proof of concept for combined pathway activation and supported further investigation of zenagamtide in individuals with type 2 diabetes to characterise its glycaemic and weight-lowering effects.

Added value of this study

To our knowledge, this was the first clinical trial to investigate the dose–response relationship of once-weekly subcutaneous zenagamtide (0.4, 1.5, 5, 10, 20, and 40 mg) in adults with type 2 diabetes, and to compare its efficacy and safety with placebo, with glycaemic control as the primary efficacy outcome. All zenagamtide doses produced clinically meaningful and significantly greater reductions in HbA_{1c} from baseline to week 36 than placebo. Improvements in continuous glucose monitoring metrics were consistent with these findings: zenagamtide increased time in range (TIR; interstitial glucose 3.9–10.0 mmol/L [70–180 mg/dL]), with significantly greater improvements than with placebo across all doses except zenagamtide 0.4 mg. The observed mean percentage of TIR at week 36 ranged from 73.8% to 91.5% (in the lowest to highest zenagamtide dose groups, equivalent to approximately 18 h per day and 22 h per day, respectively), compared with 65.3% (approximately 15 h 40 min per day) in the placebo group. Significant reductions in bodyweight compared with placebo were observed with all zenagamtide doses except 0.4 mg, with no apparent plateau at higher doses by week 36. The largest estimated percentage change in bodyweight was –14.6% with zenagamtide 40 mg at week 36. Zenagamtide showed a safety and tolerability profile consistent with GLP-1 and amylin receptor agonists. Most adverse events were gastrointestinal, mild to moderate in severity, and occurred primarily during dose escalation.

Implications of all the available evidence

The results from this phase 2 trial provide evidence of the therapeutic potential of the novel unimolecular agonist zenagamtide and support further investigation of once-weekly subcutaneous zenagamtide in phase 3 clinical trials in individuals with type 2 diabetes.

GLP-1 receptor agonists have shown positive effects on glycaemic control, bodyweight reduction, and broader cardiovascular outcomes, supporting their role in addressing the interconnected cardiometabolic risks that characterise type 2 diabetes, overweight, and obesity.^{4–9} However, attaining and sustaining optimal glycaemic control and healthy weight loss remains challenging for many individuals, highlighting the importance of individualised combination treatment strategies that target both glycated haemoglobin (HbA_{1c}) and bodyweight through complementary mechanisms of action.^{1,10}

Amylin is a neuroendocrine peptide hormone that is co-secreted with insulin from pancreatic β cells in response to food intake. Amylin helps to regulate bodyweight by increasing satiation and reducing food intake, and preclinical evidence suggests that amylin could increase energy expenditure. Amylin also improves insulin sensitivity and decreases postprandial glucagon secretion.^{11,12} Investigations of subcutaneous administration of the amylin analogues pramlintide

and cagrilintide have shown significant reductions in bodyweight and blood glucose concentrations in individuals with type 2 diabetes and overweight or obesity.^{13,14}

Previous studies have investigated the simultaneous activation of the GLP-1 and amylin pathways using a fixed-dose combination of subcutaneous cagrilintide and semaglutide in a single product (CagriSema).^{15,16} Treatment with CagriSema produced greater weight loss than treatment with either cagrilintide or semaglutide alone in adults with type 2 diabetes and a BMI of at least 27 kg/m².¹⁵ In a phase 3 trial in adults with type 2 diabetes, once-weekly CagriSema led to superior reductions in both HbA_{1c} and bodyweight compared with the monocomponents cagrilintide and semaglutide, reinforcing the metabolic benefit of combined GLP-1 and amylin pathway activation.¹⁷ Complementing these findings, a phase 3 trial extended this evidence to adults with type 2 diabetes receiving basal insulin, in which the addition of CagriSema to

background basal insulin therapy produced clinically meaningful improvements in glycaemic control and robust bodyweight loss compared with placebo, with a safety profile consistent with that of incretin-based and amylin-based therapies.¹⁸

Zenagamtide (formerly known as amycretin) is a novel unimolecular peptide agonist of GLP-1, amylin, and calcitonin receptors under investigation for weight management and treatment of type 2 diabetes. Zenagamtide has been shown in preclinical models to reduce energy intake, preserve energy expenditure, improve insulin sensitivity, and favourably affect metabolic dysfunction, supporting a mechanistic rationale for combined targeting of GLP-1 and amylin pathways in metabolic disease.^{12,19} A phase 1 first-in-human trial and a phase 1b/2a trial, both in individuals with overweight or obesity without type 2 diabetes, showed a significant weight-lowering effect of once-daily oral and once-weekly subcutaneous zenagamtide compared with placebo. In both studies, zenagamtide had a safety and tolerability profile consistent with those of GLP-1 and amylin receptor agonists.^{20,21}

This current dose-finding phase 2 trial is the first clinical evaluation of zenagamtide in individuals with type 2 diabetes. We aimed to investigate the dose-response relationship with respect to the efficacy and safety of once-weekly subcutaneous zenagamtide compared with placebo in adults with type 2 diabetes.

Methods

Study design and participants

This multicentre, randomised, parallel, double-blind (within treatment groups), placebo-controlled, dose-finding, phase 2 trial assessed the dose-response relationship of six once-weekly subcutaneous doses of zenagamtide (0.4, 1.5, 5, 10, 20, and 40 mg) and compared their effects on glycaemic control and bodyweight with placebo. The trial was conducted at 83 hospital and clinic sites across 11 countries (Bulgaria, Croatia, Germany, Greece, Hungary, Japan, Poland, Romania, Slovakia, Spain, and the USA). The trial consisted of a 3-week screening period, a 36-week intervention period, and a 4-week follow-up period (appendix p 13). A summary of the study protocol and the statistical analysis plan can be found in the appendix (pp 21–200).

Adults aged 18–75 years with type 2 diabetes (HbA_{1c} 7.0–10.0% [53–86 mmol/mol]), treated with stable doses of metformin with or without a SGLT2 inhibitor, were eligible for enrolment. Additional eligibility criteria included a BMI of 23.0 kg/m² to less than 50.0 kg/m² at screening. Per protocol, a maximum of 10% of randomly assigned participants could have a BMI below 25.0 kg/m². Participants with abnormal calcium metabolism or thyroid-calcitonin-related cancer were excluded. Full inclusion and exclusion criteria are detailed in the appendix (pp 3–4). Participants were

asked to report their biological sex (male or female) by trial personnel; their responses were collected in an electronic data capture system.

The trial was conducted in compliance with the principles of the Declaration of Helsinki and in accordance with the Good Clinical Practice guidelines of the International Conference for Harmonisation.^{22,23} The protocol, consent form, and other relevant documents were reviewed and approved by the appropriate institutional review boards or independent ethics committees in each country (appendix p 5). All participants provided written informed consent before trial entry. There was no patient or public involvement in the design, conduct, and reporting of the trial. This trial was registered with ClinicalTrials.gov (NCT06542874) on Aug 5, 2024, and has been completed.

Randomisation and masking

A randomisation and trial supplies management system was used to assign eligible participants (7:5) to the once-weekly subcutaneous or the once-daily oral zenagamtide treatment groups. Details of further randomisation for the once-daily oral treatment group are reported separately.²⁴ Participants assigned to the subcutaneous treatment group were randomly assigned (6:1) to subcutaneous zenagamtide or placebo. Participants assigned to subcutaneous zenagamtide were then randomly assigned (1:1:1:1:1:1) to one of six maintenance-dose groups (0.4, 1.5, 5, 10, 20, and 40 mg). Participants assigned to placebo were similarly allocated to matched placebo regimens corresponding to each dosing schedule; individual placebo groups were pooled for analysis. This strategy planned for 36 participants to receive active treatment and six to receive placebo in each of the groups. Randomisation was stratified by HbA_{1c} (<8.5% or ≥8.5%) at screening and country of residence (Japan or other countries). The overall trial population aimed to include approximately 10% participants from Japan to support their inclusion in phase 3 trials. The trial was double-blind within each dose level, with participants and investigators masked to allocation to zenagamtide or matched placebo. Participants and investigators were not masked to the assigned dose level.

See Online for appendix

Procedures

Subcutaneous zenagamtide and matched placebo were administered in an identical manner via a pen device (NovoPen 4, Novo Nordisk, Søborg, Denmark) once a week on the same day each week, at any time of the day. The starting dose of subcutaneous zenagamtide was 0.2 mg, with dose escalations every 4 weeks until the maintenance dose was reached (0.4–40 mg). Depending on the maintenance dose level, the maintenance period ranged from 4 weeks to 32 weeks, according to the dose-escalation schedule, up to a total treatment period of 36 weeks (appendix p 13). Dose modification from the randomly assigned dose was not allowed within the

fixed-dose-escalation trial design. Therefore, if the planned dose was not tolerated, the participant permanently discontinued trial treatment. Randomly assigned treatments were continued until week 36 or until the participant permanently discontinued treatment. In cases of persistent hyperglycaemia, rescue medication (intensification or initiation of glucose-lowering medications, excluding GLP-1 receptor agonists, DPP-4 inhibitors, GLP-1 and GIP receptor dual agonists, and amylin analogues) was prescribed to participants at the investigators' discretion. Blood glucose concentrations were measured using the provided blood glucose meter (Abbott Freestyle Optium Neo, Abbott, Berkshire, UK) when a hypoglycaemic episode was suspected. Participants were provided with a continuous glucose monitoring (CGM) device (Dexcom G6, Dexcom, San Diego, CA, USA). CGM data were collected during three prespecified measurement periods of approximately 10 days starting at screening (used as baseline), week 18, and week 34. Both the participants and investigators were masked to CGM data, and these data were not used for any dose adjustments or reporting of hypoglycaemic events.

Outcomes

The primary outcome was change in HbA_{1c} from baseline to week 36. Supportive (ie, to reinforce or provide context for the primary outcome) secondary outcomes were changes from baseline to week 36 in relative and absolute bodyweight; fasting plasma glucose (FPG); BMI; waist circumference; systolic blood pressure (SBP) and 24 h SBP; high-sensitivity C-reactive protein (hsCRP); total cholesterol, HDL cholesterol, LDL cholesterol, and triglycerides; urinary albumin-to-creatinine ratio (UACR); and estimated glomerular filtration rate (eGFR) based on the Chronic Kidney Disease Epidemiology Collaboration (Creatinine-Cystatin C) equation (2021).²⁵ Other secondary outcomes included the number of participants without macroalbuminuria (UACR <300 mg/g) at baseline developing new-onset macroalbuminuria (UACR ≥300 mg/g) at week 36, change in percentage of time in range (TIR; interstitial glucose 3.9–10.0 mmol/L [70–180 mg/dL]) from baseline to weeks 20 and 36, and the number of adverse events from baseline (week 0) to end of trial (week 40). Prespecified exploratory outcomes included attainment of HbA_{1c} targets (<7%, ≤6.5%, and <5.7%) and weight loss targets (≥5%, ≥10%, ≥15%, ≥20%, and ≥25% weight reduction) at week 36, and change in pulse from baseline to week 36. Further outcomes reported alongside the prespecified study outcomes included the percentage of TIR, time in tight range (T1TR; 3.9–7.8 mmol/L [70–140 mg/dL]), and time below range (TBR; <3.9 mmol/L [70 mg/dL] and <3.0 mmol/L [54 mg/dL]).

Statistical analysis

The primary objective of this trial was to demonstrate and characterise the dose–response relationship of

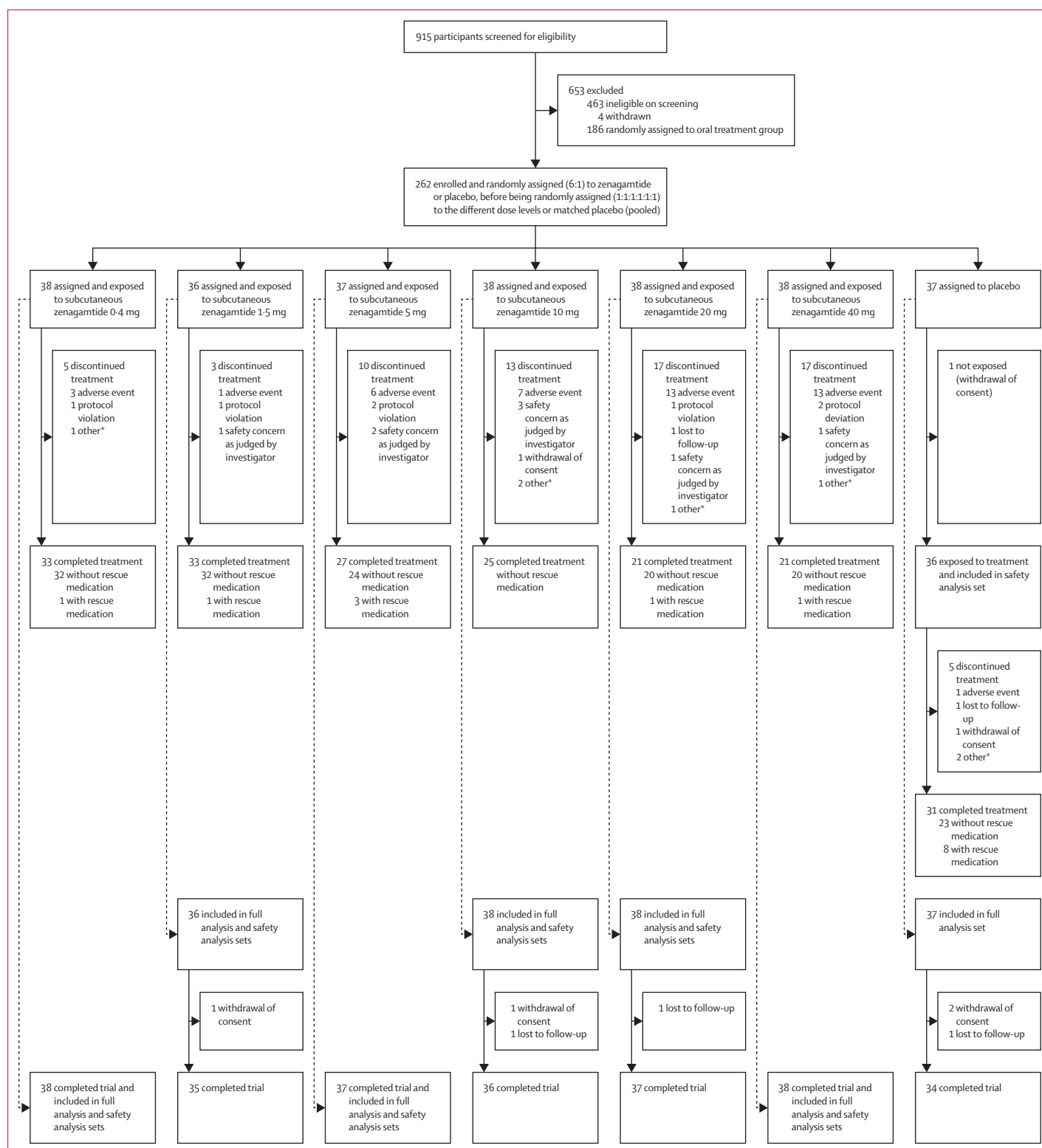
once-weekly subcutaneous zenagamtide, and to compare the effect of varying doses with that of placebo, for change in HbA_{1c} from baseline to week 36. Sample size calculations were based on simulations using a maximum effect ($E_{\max}$) model for the primary outcome, assuming that the baseline effect was 0, the median effective dose (ED_{50}) was 0.6 mg for subcutaneous zenagamtide, $E_{\max}$ was –2.2%, and that the SD was 1.1%. The SD in this trial was based on results from a previous trial with CagriSema (cagrilintide 2.4 mg and semaglutide 2.4 mg) in individuals with type 2 diabetes, in which the SD was 0.9% after 32 weeks of treatment.¹⁵ The $E_{\max}$ and ED_{50} assumptions were based on a phase 1 trial of zenagamtide.²¹ Based on these assumptions, it was considered that the random assignment of 36 participants to each active group would ensure sufficient precision to characterise a dose–response relationship for HbA_{1c}.

The full analysis set included all randomly assigned participants, whereas the safety analysis set included all participants exposed to treatment. The efficacy estimand was estimated based on the full analysis set using data collected while participants remained on trial treatment without initiation of rescue medication (appendix p 6). Imputation of missing data was handled by multiple imputation, assuming that missing data were missing at random.

An ANCOVA model with randomly assigned treatment and HbA_{1c} stratification (<8.5% vs ≥8.5%) as fixed factors and baseline HbA_{1c} as a fixed covariate was used to compare the effect of varying zenagamtide doses with placebo for change in HbA_{1c} from baseline to week 36. A similar approach was used to evaluate the relative change in bodyweight, with baseline bodyweight used as a fixed covariate. Change in BMI, eGFR, FPG, hsCRP, lipid parameters, 24 h SBP, SBP, TIR, UACR, and waist circumference from baseline to week 36 were analysed using an ANCOVA model with the outcome as a dependent variable, randomly assigned treatment and strata as fixed factors, and the baseline assessment corresponding to the outcome as a fixed covariate. Attainment of HbA_{1c} and weight loss targets at week 36 were analysed using a logistic regression model, with the outcome as a dependent variable, randomised treatment and strata as fixed factors, and the baseline assessment corresponding to the outcome as a fixed covariate. All statistical analyses were prespecified. *p* values are nominal and were not adjusted for multiplicity; all results should be interpreted as exploratory or descriptive. Imputation was performed using SAS (version 9.4), and all other statistical analyses were conducted using R (version 4.0.4).

Role of the funding source

Representatives of Novo Nordisk were involved in the trial design and conduct; data collection, management, analysis, and interpretation; and preparation, review, and approval of the manuscript. Novo Nordisk did not have the right to veto publication or to control the decision



	Zenagamtide 0.4 mg (n=38)	Zenagamtide 1.5 mg (n=36)	Zenagamtide 5 mg (n=37)	Zenagamtide 10 mg (n=38)	Zenagamtide 20 mg (n=38)	Zenagamtide 40 mg (n=38)	Pooled placebo (n=37)
Sex							
Male	23 (61%)	23 (64%)	23 (62%)	26 (68%)	26 (68%)	26 (68%)	26 (70%)
Female	15 (39%)	13 (36%)	14 (38%)	12 (32%)	12 (32%)	12 (32%)	11 (30%)
Age, years	56.0 (10.4)	56.9 (8.8)	57.0 (10.2)	57.6 (9.1)	58.0 (9.3)	57.2 (9.0)	57.2 (10.6)
HbA _{1c} , %	8.0 (0.8)	7.7 (0.9)	7.8 (0.7)	7.9 (0.8)	7.7 (0.6)	7.8 (0.8)	8.0 (0.7)
HbA _{1c} , mmol/mol	64.4 (9.0)	60.6 (9.5)	61.6 (7.6)	62.3 (8.7)	61.0 (7.0)	62.1 (8.6)	63.5 (7.2)
HbA _{1c} strata							
<8.5%	28 (74%)	29 (81%)	32 (87%)	32 (84%)	36 (95%)	31 (82%)	27 (73%)
≥8.5%	10 (26%)	7 (19%)	5 (14%)	6 (16%)	2 (5%)	7 (18%)	10 (27%)
FPG, mmol/L	8.6 (2.2)	8.6 (2.3)	9.1 (1.8)	8.3 (1.8)	8.2 (1.9)	8.9 (2.5)	8.9 (2.2)
FPG, mg/dL	155 (40)	155 (42)	164 (32)	149 (33)	148 (34)	161 (44)	159 (40)
Diabetes duration, years	8 (8)	8 (5)	10 (11)	9 (7)	6 (6)	8 (7)	8 (6)
Bodyweight, kg	102.4 (25.2)	99.9 (20.2)	98.9 (20.1)	95.8 (21.0)	98.4 (21.0)	99.5 (25.2)	99.8 (22.8)
BMI, kg/m ²	35.3 (6.6)	34.8 (6.7)	33.7 (6.2)	32.9 (6.4)	32.6 (5.4)	32.9 (6.1)	33.9 (5.9)
Waist circumference, cm	115.7 (16.9)	114.6 (15.3)	112.7 (12.5)	108.3 (14.4)	108.5 (12.7)	112.2 (15.1)	112.2 (13.3)
Systolic blood pressure, mm Hg	131.1 (14.7)	133.6 (13.6)	133.2 (13.9)	129.4 (13.8)	129.2 (12.7)	130.4 (13.8)	129.9 (13.8)
Diastolic blood pressure, mm Hg	80.9 (7.1)	80.0 (9.2)	82.8 (8.4)	79.9 (8.1)	80.7 (9.7)	82.3 (8.0)	80.2 (8.5)
eGFR, * mL/min per 1.73 m ²	88 (20)	88 (19)	92 (16)	90 (19)	88 (19)	84 (22)	87 (21)
SGLT2 inhibitor use at randomisation	16 (42%)	10 (28%)	16 (43%)	14 (37%)	16 (42%)	16 (42%)	16 (43%)
Race							
White	34 (89%)	31 (86%)	32 (86%)	34 (89%)	34 (89%)	33 (87%)	34 (92%)
Asian	4 (11%)	3 (8%)	4 (11%)	4 (11%)	4 (11%)	4 (11%)	3 (8%)
Black or African American	0	1 (3%)	1 (3%)	0	0	1 (3%)	0
American Indian or Alaskan native†	0	1 (3%)	0	0	0	0	0

Data are n (%) or mean (SD). The full analysis set included all randomly assigned participants. CKD-EPI=Chronic Kidney Disease Epidemiology Collaboration. eGFR=estimated glomerular filtration rate. FPG=fasting plasma glucose. *eGFR calculated using the CKD-EPI (Creatinine-Cystatin C) equation (2021).²⁵ †Self-reported by participants.

Table 1: Participant demographics and baseline characteristics (full analysis set)

regarding journal submission. Additionally, Novo Nordisk funded medical writing support to aid with the drafting of the manuscript under the direction of the authors.

Results

Participants were recruited between Aug 7 and Dec 27, 2024. Of the 915 individuals who were screened, 262 were randomly assigned to the subcutaneous treatment group and 186 were randomly assigned to the oral treatment group. Details of ineligibility on screening are reported in the appendix (pp 7–8). Details of the oral treatment group are reported separately.²⁴ The 262 participants in the subcutaneous treatment group were randomly assigned to receive subcutaneous zenagamtide (n=225) or placebo (n=37); 261 participants were exposed to treatment (figure 1). Baseline characteristics are shown in table 1. At baseline, participants had a mean age of 57.1 years (SD 9.6), HbA_{1c} of 7.8% (0.8), BMI of 33.7 kg/m² (6.2), and diabetes duration of 8 years (7). 173 (66%) of 262 participants were male. In total, 191 (73%) of 262 participants completed trial treatment. Treatment discontinuation ranged from three (8%) of 36 participants in the zenagamtide 1.5 mg group to 17 (45%) of 38 participants in the zenagamtide 20 mg and 40 mg

groups. In the placebo group, five (14%) of 36 participants discontinued treatment. Most treatment discontinuations were due to adverse events (44 [17%] of 262 participants), with the largest numbers of discontinuations reported in the groups assigned to higher doses of zenagamtide (figure 1). Discontinuations primarily occurred during dose escalations (appendix p 14).

Rescue medication was required by one participant in each of the zenagamtide 0.4 mg, 1.5 mg, 20 mg, and 40 mg groups, three participants in the zenagamtide 5 mg group, and eight participants in the placebo group. Rescue medication was not required in any of the participants in the zenagamtide 10 mg group (figure 1).

Unless otherwise stated, efficacy results are estimated values based on the efficacy estimand. Zenagamtide showed significantly greater reductions in HbA_{1c} from baseline to week 36 than placebo at all doses (figure 2A), with a significant dose–response relationship (appendix p 9). Estimated change in HbA_{1c} from a mean of 7.8% (SD 0.8) at baseline across groups to week 36 ranged from –0.9% with zenagamtide 0.4 mg (estimated treatment difference [ETD] vs placebo –0.77% [95% CI –1.26 to –0.28]; p=0.0021) to –1.7% with zenagamtide 40 mg (–1.56% [–2.05 to –1.07]; p<0.0001; table 2).

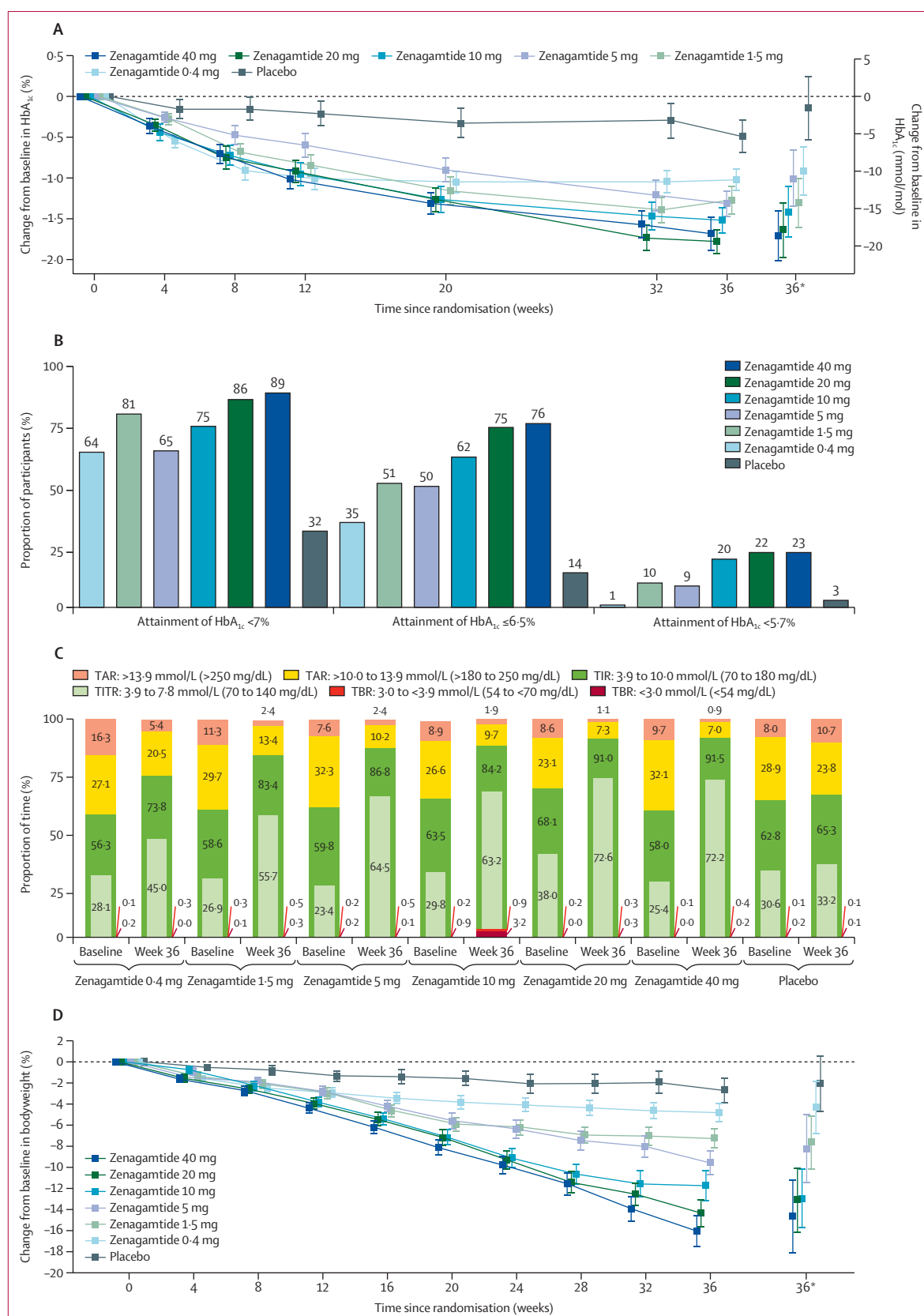


Figure 2: Key efficacy outcomes

(A) Mean change in HbA_{1c} from baseline to week 36. Error bars show SE of the mean. Estimated values based on the efficacy estimand, incorporating the full analysis set (all randomly assigned participants) and the on-treatment without rescue medication datapoint set. (B) Proportion of participants attaining HbA_{1c} targets at week 36. Each binary responder outcome was analysed using a logistic regression model with the outcome as the dependent variable and with randomly assigned treatment and strata as fixed-effect factors and baseline of the underlying parameter as a fixed-effect covariate. Missing data were imputed using multiple imputation. (C) CGM metrics at baseline and week 36. Outcomes assessed as observed means. CGM assessments are averages calculated over a 10-day period. Time spent is defined as the proportion of time spent in a given range. (D) Mean change in bodyweight from baseline to week 36. Error bars show SE of the mean. Estimated values based on the efficacy estimand, incorporating the full analysis set (all randomly assigned participants) and the on-treatment without rescue medication datapoint set. Missing data were imputed using multiple imputation and data from the on-treatment without rescue medication observation period. CGM=continuous glucose monitoring. TAR=time above range. TBR=time below range. TIR=time in range. TITR=time in tight range. *Data shown at week 36 are the estimated mean values and the corresponding SE from statistical analysis.

	Zenagamtide 0.4 mg (n=38)	Zenagamtide 1.5 mg (n=36)	Zenagamtide 5 mg (n=37)	Zenagamtide 10 mg (n=38)	Zenagamtide 20 mg (n=38)	Zenagamtide 40 mg (n=38)	Pooled placebo (n=37)
HbA_{1c} %							
Baseline	8.0 (0.8)	7.7 (0.9)	7.8 (0.7)	7.9 (0.8)	7.7 (0.6)	7.8 (0.8)	8.0 (0.7)
Change from baseline at week 36	-0.9	-1.3	-1.0	-1.4	-1.6	-1.7	-0.1
ETD vs placebo (95% CI); p value	-0.77 (-1.26 to -0.28); 0.0021	-1.17 (-1.66 to -0.67); <0.0001	-0.86 (-1.38 to -0.34); 0.0013	-1.27 (-1.77 to -0.77); <0.0001	-1.49 (-2.00 to -0.98); <0.0001	-1.56 (-2.05 to -1.07); <0.0001	..
TIR 3.9–10.0 mmol/L (70–180 mg/dL), %							
Baseline	56.3 (33.3)	58.6 (26.1)	59.8 (20.9)	63.5 (22.9)	68.1 (25.3)	58.0 (24.6)	62.8 (26.5)
Change from baseline at week 20	20.8	25.9	27.8	26.9	20.9	29.9	9.8
ETD vs placebo (95% CI); p value	11.02 (1.93 to 20.10); 0.018	16.09 (6.61 to 25.57); 0.0010	18.05 (9.03 to 27.07); 0.0001	17.15 (7.99 to 26.30); 0.0003	11.09 (1.23 to 20.95); 0.028	20.10 (10.97 to 29.24); <0.0001	..
Change from baseline at week 36	13.9	23.7	24.8	23.7	24.0	30.4	-0.4
ETD vs placebo (95% CI); p value	14.37 (-0.08 to 28.83); 0.051	24.11 (9.62 to 38.60); 0.0013	25.25 (10.59 to 39.90); 0.0009	24.15 (9.18 to 39.11); 0.0019	24.41 (9.65 to 39.18); 0.0014	30.85 (16.74 to 44.95); <0.0001	..
FPG, mmol/L							
Baseline	8.6 (2.2)	8.6 (2.3)	9.1 (1.8)	8.3 (1.8)	8.2 (1.9)	8.9 (2.5)	8.9 (2.2)
Change from baseline at week 36	-1.3	-2.2	-2.1	-2.3	-2.4	-2.8	0.3
ETD vs placebo (95% CI); p value	-1.68 (-2.95 to -0.42); 0.0094	-2.54 (-3.81 to -1.26); 0.0001	-2.41 (-3.90 to -0.91); 0.0020	-2.63 (-3.93 to -1.33); 0.0001	-2.73 (-4.02 to -1.44) p<0.0001	-3.11 (-4.39 to -1.82) p<0.0001	..
FPG, mg/dL							
Baseline	155.1 (39.5)	154.8 (41.7)	164.1 (31.7)	149.2 (33.2)	148.3 (33.8)	160.6 (44.2)	159.4 (39.6)
Change from baseline at week 36	-24.1	-39.5	-37.1	-41.1	-43.0	-49.8	6.2
ETD vs placebo (95% CI); p value	-30.34 (-53.09 to -7.59); 0.0094	-45.72 (-68.71 to -22.73); 0.0001	-43.37 (-70.35 to -16.39); 0.0020	-47.37 (-70.86 to -23.88); 0.0001	-49.24 (-72.51 to -25.97); <0.0001	-55.98 (-79.17 to -32.79); <0.0001	..
Bodyweight change, %							
Baseline, kg	102.4 (25.2)	99.9 (20.2)	98.9 (20.1)	95.8 (21.0)	98.4 (21.0)	99.5 (25.2)	99.8 (22.8)
Change from baseline at week 36	-4.3	-7.6	-8.2	-12.9	-13.1	-14.6	-2.1
ETD vs placebo (95% CI); p value	-2.25 (-5.83 to 1.34); 0.22	-5.52 (-9.15 to -1.90); 0.0030	-6.09 (-10.20 to -1.99); 0.0039	-10.82 (-14.61 to -7.03); <0.0001	-11.04 (-15.04 to -7.03); <0.0001	-12.50 (-16.79 to -8.21); <0.0001	..
Bodyweight, kg							
Baseline	102.4 (25.2)	99.9 (20.2)	98.9 (20.1)	95.8 (21.0)	98.4 (21.0)	99.5 (25.2)	99.8 (22.8)
Change from baseline at week 36	-4.0	-7.5	-8.4	-12.1	-12.3	-14.9	-1.8
ETD vs placebo (95% CI); p value	-2.23 (-5.68 to 1.22); 0.20	-5.72 (-9.22 to -2.23); 0.0014	-6.63 (-10.43 to -2.84); 0.0007	-10.30 (-13.90 to -6.71); <0.0001	-10.51 (-14.40 to -6.63); <0.0001	-13.11 (-17.14 to -9.08); <0.0001	..
BMI, kg/m²							
Baseline	35.3 (6.6)	34.8 (6.7)	33.7 (6.2)	32.9 (6.4)	32.6 (5.4)	32.9 (6.1)	33.9 (5.9)
Change from baseline at week 36	-1.4	-2.6	-2.9	-4.2	-4.2	-5.0	-0.6
ETD vs placebo (95% CI); p value	-0.74 (-1.92 to 0.44); 0.22	-1.96 (-3.15 to -0.76); 0.0015	-2.25 (-3.54 to -0.95); 0.0008	-3.57 (-4.80 to -2.33); <0.0001	-3.60 (-4.95 to -2.26); <0.0001	-4.33 (-5.70 to -2.96); <0.0001	..
Waist circumference, cm							
Baseline	115.7 (16.9)	114.6 (15.3)	112.7 (12.5)	108.3 (14.4)	108.5 (12.7)	112.2 (15.1)	122.2 (13.3)
Change from baseline at week 36	-3.2	-6.5	-6.5	-9.9	-11.5	-11.7	-0.9
ETD vs placebo (95% CI); p value	-2.28 (-5.62 to 1.06); 0.18	-5.60 (-8.96 to -2.23); 0.0012	-5.61 (-9.22 to -2.00); 0.0025	-8.98 (-12.50 to -5.46); <0.0001	-10.58 (-14.36 to -6.81); <0.0001	-10.76 (-15.96 to -5.56); <0.0001	..

(Table 2 continues on next page)

A higher proportion of participants attained HbA_{1c} targets (<7·0% and ≤6·5%) with zenagamtide than with placebo across all doses (64–89% of participants attained HbA_{1c} <7·0% and 35–76% attained HbA_{1c} ≤6·5% across the lowest to highest zenagamtide doses, respectively,

compared with 32% and 14%, respectively, in the pooled placebo group; figure 2B). Similarly, a higher proportion of participants attained HbA_{1c} below 5·7% with all zenagamtide doses except 0·4 mg than with placebo (figure 2B).

	Zenagamtide 0·4 mg (n=38)	Zenagamtide 1·5 mg (n=36)	Zenagamtide 5 mg (n=37)	Zenagamtide 10 mg (n=38)	Zenagamtide 20 mg (n=38)	Zenagamtide 40 mg (n=38)	Pooled placebo (n=37)
(Continued from previous page)							
hsCRP, mg/L							
Baseline	4·7 (5·2)	4·0 (4·4)	6·0 (10·5)	3·8 (4·9)	3·2 (3·6)	4·2 (4·1)	3·5 (4·2)
Ratio to baseline at week 36	0·8	0·7	0·6	0·5	0·6	0·3	1·2
ETR vs placebo (95% CI); p value	0·71 (0·27 to 1·87); 0·48	0·60 (0·23 to 1·59); 0·30	0·52 (0·19 to 1·38); 0·19	0·41 (0·15 to 1·10); 0·077	0·54 (0·17 to 1·69); 0·29	0·26 (0·09 to 0·76); 0·015	..
HDL cholesterol, mmol/L							
Baseline	1·1 (0·3)	1·2 (0·3)	1·1 (0·2)	1·1 (0·3)	1·2 (0·3)	1·1 (0·3)	1·1 (0·2)
Ratio to baseline at week 36	1·1	1·0	1·1	1·0	1·1	1·1	1·0
ETR vs placebo (95% CI)	1·06 (0·97 to 1·14); 0·19	1·04 (0·96 to 1·13); 0·33	1·07 (0·99 to 1·17); 0·099	1·05 (0·97 to 1·15); 0·23	1·09 (1·00 to 1·19); 0·047	1·14 (1·04 to 1·24); 0·0044	..
LDL cholesterol, mmol/L							
Baseline	2·7 (0·8)	2·7 (0·9)	2·6 (0·9)	2·3 (0·9)	2·5 (1·0)	2·5 (0·9)	2·5 (0·8)
Ratio to baseline at week 36	0·9	0·9	0·7	0·8	0·8	0·8	0·8
ETR vs placebo (95% CI); p value	1·15 (0·94 to 1·40); 0·17	1·19 (0·97 to 1·45); 0·090	0·95 (0·76 to 1·18); 0·62	1·05 (0·85 to 1·32); 0·64	1·07 (0·87 to 1·33); 0·52	1·02 (0·82 to 1·26); 0·86	..
Total cholesterol, mmol/L							
Baseline	4·7 (0·9)	4·8 (1·1)	4·6 (1·0)	4·3 (1·0)	4·5 (1·2)	4·4 (1·0)	4·5 (0·9)
Ratio to baseline at week 36	0·9	0·9	0·9	0·9	0·9	0·8	1·0
ETR vs placebo (95% CI); p value	0·98 (0·86 to 1·12); 0·75	0·97 (0·85 to 1·11); 0·66	0·91 (0·79 to 1·04); 0·17	0·91 (0·78 to 1·05); 0·18	0·88 (0·77 to 1·02); 0·085	0·88 (0·76 to 1·01); 0·074	..
Triglycerides, mmol/L							
Baseline	2·0 (1·0)	2·0 (1·0)	2·0 (0·9)	2·0 (1·1)	2·0 (1·2)	2·0 (0·9)	2·0 (0·9)
Ratio to baseline at week 36	0·9	0·8	0·8	0·7	0·6	0·6	1·1
ETR vs placebo (95% CI); p value	0·80 (0·60 to 1·06); 0·12	0·75 (0·56 to 1·00); 0·048	0·76 (0·57 to 1·03); 0·076	0·63 (0·47 to 0·84); 0·0022	0·57 (0·43 to 0·76); 0·0002	0·52 (0·38 to 0·71); <0·0001	..
SBP, mm Hg							
Baseline	131·1 (14·7)	133·6 (13·6)	133·2 (13·9)	129·4 (13·8)	129·2 (12·7)	130·4 (13·8)	129·9 (13·8)
Change from baseline at week 36	–5·8	–3·4	–7·7	–9·4	–7·2	–10·6	–3·5
ETD vs placebo (95% CI); p value	–2·28 (–11·10 to 6·53); 0·61	0·09 (–8·82 to 9·00); 0·98	–4·17 (–13·68 to 5·34); 0·39	–5·87 (–14·97 to 3·24); 0·21	–3·70 (–14·27 to 6·87); 0·49	–7·13 (–19·24 to 4·98); 0·25	..
24 h SBP, mm Hg							
Baseline	130·2 (16·7)	129·2 (9·5)	131·7 (11·1)	126·1 (10·0)	128·9 (15·5)	129·4 (13·4)	127·4 (11·4)
Change from baseline at week 36	–3·9	–4·0	–6·6	–6·7	–10·1	–9·9	–0·9
ETD vs placebo (95% CI); p value	–3·05 (–9·73 to 3·64); 0·37	–3·12 (–9·61 to 3·37); 0·34	–5·76 (–12·55 to 1·03); 0·096	–5·85 (–13·76 to 2·06); 0·15	–9·20 (–16·44 to –1·96); 0·013	–9·06 (–17·57 to –0·56); 0·037	..
eGFR, * mL/min per 1·73 m²							
Baseline	88 (20)	88 (19)	92 (16)	90 (19)	88 (19)	84 (22)	87 (21)
Change from baseline at week 36	3	1	6	11	8	8	–2
ETD vs placebo (95% CI); p value	4·25 (–1·85 to 10·35); 0·17	2·84 (–3·30 to 8·99); 0·36	7·42 (0·71 to 14·13); 0·031	12·59 (6·27 to 18·92); 0·0001	9·55 (2·64 to 16·46); 0·0072	9·69 (2·97 to 16·40); 0·0050	..

(Table 2 continues on next page)

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UACR, mg/g

	Zenagamtide 0.4 mg (n=38)	Zenagamtide 1.5 mg (n=36)	Zenagamtide 5 mg (n=37)	Zenagamtide 10 mg (n=38)	Zenagamtide 20 mg (n=38)	Zenagamtide 40 mg (n=38)	Pooled placebo (n=37)
Baseline†	19.7 (107.9)	25.7 (226.8)	19.7 (98.4)	16.3 (119.9)	20.9 (113.8)	15.8 (126.2)	28.0 (163.4)
Ratio to baseline at week 36	0.6	0.8	0.6	0.8	0.5	0.7	1.2
ETR vs placebo (95% CI); p value	0.51 (0.26 to 1.00); 0.051	0.62 (0.31 to 1.24); 0.18	0.44 (0.21 to 0.92); 0.029	0.68 (0.32 to 1.46); 0.32	0.39 (0.19 to 0.80); 0.012	0.58 (0.27 to 1.25); 0.16	..

The full analysis set included all randomly assigned participants. Data at baseline are observed values and are shown as mean (SD), unless otherwise stated. All other data are estimated values based on the efficacy estimand, incorporating the full analysis set and the on-treatment without rescue medication datapoint set. CKD-EPI=Chronic Kidney Disease Epidemiology Collaboration. eGFR=estimated glomerular filtration rate. ETD=estimated treatment difference. ETR=estimated treatment ratio. FPG=fasting plasma glucose. hsCRP=high-sensitivity C-reactive protein. SBP=systolic blood pressure. TIR=time in range. UACR=urinary albumin-to-creatinine ratio. *eGFR calculated using the CKD-EPI (Creatinine-Cystatin C) equation (2021).²⁵ †Geometric mean (coefficient of variation).

Table 2: Summary of key outcomes (full analysis set)

From baseline to week 20, the observed percentage of TIR (3.9–10.0 mmol/L [70–180 mg/dL]) increased with all zenagamtide doses, as did the percentage of T1TR (3.9–7.8 mmol/L [70–140 mg/dL]; appendix p 15). Further increases in observed percentages of TIR and T1TR were seen from week 20 to week 36 in the 20 mg and 40 mg groups as the participants reached their maintenance dose (figure 2C; appendix p 15). The observed percentage of TIR at week 36 ranged from 73.8% (equivalent to approximately 18 h per day) in the lowest dose group to 91.5% (approximately 22 h per day) in the highest zenagamtide dose group, compared with 65.3% (approximately 15 h 40 min per day) in the pooled placebo group. The observed percentage of T1TR at week 36 ranged from 45.0% (approximately 11 h per day) in the lowest dose group to 72.6% (approximately 17 h 25 min per day) in the zenagamtide 20 mg group, and was 33.2% (approximately 8 h per day) in the placebo group (figure 2C). Estimated change in percentage of TIR from baseline to week 36 ranged from 13.9% with zenagamtide 0.4 mg (ETD vs placebo 14.37 [95% CI –0.08 to 28.83]) to 30.4% with zenagamtide 40 mg (30.85 [16.74 to 44.95]; table 2). The estimated increase in percentage of TIR from baseline to week 36 was significantly greater with zenagamtide than with placebo across all doses except 0.4 mg (table 2). The observed percentage of TBR below 3.9 mmol/L (<70 mg/dL) and TBR below 3.0 mmol/L (<54 mg/dL) ranged from 0.3% to 4.1% and from 0 to 3.2%, respectively, across all zenagamtide doses at week 36 (figure 2C).

FPG decreased from baseline to week 36 in all zenagamtide treatment groups (table 2). Estimated changes ranged from –1.3 mmol/L (–24.1 mg/dL) with zenagamtide 0.4 mg (ETD vs placebo: –1.68 mmol/L [95% CI –2.95 to –0.42]; –30.34 mg/dL [–53.09 to –7.59]; p=0.0094) to –2.8 mmol/L (49.8 mg/dL) with zenagamtide 40 mg (–3.11 mmol/L [–4.39 to –1.82]; –55.98 mg/dL [–79.17 to –32.79]; p<0.0001). In the placebo group, FPG increased by 0.4 mmol/L (6.2 mg/dL) from baseline to week 36 (table 2).

Reductions in bodyweight, BMI, and waist circumference were observed with zenagamtide treatment over the course of the trial and were broadly dose-dependent (table 2; appendix pp 9, 16). From baseline to week 36, significant decreases in bodyweight compared with placebo were seen with all zenagamtide doses except 0.4 mg (figure 2D; table 2). Estimated percentage change in bodyweight ranged from –4.3% with zenagamtide 0.4 mg (ETD vs placebo –2.25% [95% CI –5.83 to 1.34]; p=0.22) to –14.6% with zenagamtide 40 mg (–12.5% [–16.79 to –8.21]; p<0.0001), compared with –2.1% with placebo (table 2). No apparent weight loss plateau was seen at week 36 with higher doses of zenagamtide (5–40 mg; figure 2D). In general, a greater proportion of participants attained predefined weight loss targets (≥5%, ≥10%, ≥15%, ≥20%, and ≥25% weight reduction) with zenagamtide than with placebo (appendix p 10). Significant decreases in BMI compared with placebo were observed with all zenagamtide doses except 0.4 mg (p≤0.0015 for 1.5–40 mg; table 2). Significant reductions in waist circumference versus placebo were also observed with all zenagamtide doses except 0.4 mg (p≤0.0025 for 1.5–40 mg; table 2).

Improvements compared with placebo were also observed for SBP, hsCRP, and some lipid parameters at week 36, but few were significant (table 2; appendix p 17). The change in SBP from baseline to week 36 ranged from –3.4 mm Hg to –10.6 mm Hg with zenagamtide compared with –3.5 mm Hg with placebo. Improvements in SBP were not significant for any dose of zenagamtide versus placebo (table 2); however, there was a significant improvement in mean 24 h SBP with zenagamtide 20 mg and 40 mg compared with placebo (table 2). At week 36, the change from baseline in pulse ranged from 1.96 bpm to 8.65 bpm with all zenagamtide doses compared with –1.40 bpm with placebo (appendix pp 11, 18). From baseline to week 36, a significant increase in pulse rate was observed for the 5 mg zenagamtide dose group compared with placebo (ETD vs placebo 7.84 bpm [95% CI 0.13 to 15.56]; p=0.046); no statistically

	Zenagamtide 0·4 mg (n=38)	Zenagamtide 1·5 mg (n=36)	Zenagamtide 5 mg (n=37)	Zenagamtide 10 mg (n=38)	Zenagamtide 20 mg (n=38)	Zenagamtide 40 mg (n=38)	Pooled placebo (n=36)
Adverse events							
Total adverse events	26 (68%)	24 (67%)	27 (73%)	26 (68%)	34 (90%)	34 (90%)	21 (58%)
Serious adverse events	4 (11%)	3 (8%)	2 (5%)	5 (13%)	3 (8%)	1 (3%)	3 (8%)
Severe adverse events	2 (5%)	2 (6%)	0	5 (13%)	3 (8%)	4 (11%)	0
Moderate adverse events	7 (18%)	11 (31%)	14 (38%)	14 (37%)	18 (47%)	22 (58%)	8 (22%)
Mild adverse events	21 (55%)	18 (50%)	23 (62%)	23 (61%)	31 (82%)	30 (79%)	19 (53%)
Adverse events probably related to trial product	9 (24%)	4 (11%)	12 (32%)	16 (42%)	23 (61%)	27 (71%)	7 (19%)
Adverse events possibly related to trial product	4 (11%)	12 (33%)	13 (35%)	14 (37%)	15 (40%)	14 (37%)	5 (14%)
Outcome: not recovered*	9 (24%)	5 (14%)	9 (24%)	6 (16%)	10 (26%)	12 (32%)	8 (22%)
Outcome: fatal	0	0	0	0	0	0	0
Action taken: trial treatment discontinued	3 (8%)	1 (3%)	6 (16%)	8 (21%)	13 (34%)	13 (34%)	2 (6%)
Adverse events reported by ≥5% of participants†							
Nausea	4 (11%)	6 (17%)	14 (38%)	10 (26%)	19 (50%)	14 (37%)	5 (14%)
Vomiting	1 (3%)	1 (3%)	6 (16%)	4 (11%)	9 (24%)	11 (29%)	2 (6%)
Diarrhoea	2 (5%)	2 (6%)	4 (11%)	4 (11%)	6 (16%)	14 (37%)	2 (6%)
Constipation	2 (5%)	1 (3%)	2 (5%)	2 (5%)	3 (8%)	4 (11%)	1 (3%)
Dyspepsia	0	1 (3%)	2 (5%)	2 (5%)	3 (8%)	3 (8%)	0
Abdominal discomfort	0	1 (3%)	1 (3%)	3 (8%)	2 (5%)	3 (8%)	0
Nasopharyngitis	0	1 (3%)	2 (5%)	1 (3%)	6 (16%)	7 (18%)	3 (8%)
Influenza	1 (3%)	1 (3%)	1 (3%)	2 (5%)	0	2 (5%)	2 (6%)
Lipase increased	2 (5%)	2 (6%)	5 (14%)	0	3 (8%)	3 (8%)	0
Headache	0	1 (3%)	2 (5%)	1 (3%)	3 (8%)	0	0
Decreased appetite	0	0	2 (5%)	3 (8%)	1 (3%)	6 (16%)	0
Adverse events of special interest‡							
Gastrointestinal adverse events§	9 (24%)	11 (31%)	21 (57%)	18 (47%)	24 (63%)	28 (74%)	9 (25%)
Dysaesthesia§	0	0	0	3 (8%)	2 (5%)	8 (21%)	0
Cardiovascular disorders	5 (13%)	4 (11%)	1 (3%)	2 (5%)	6 (16%)	3 (8%)	3 (8%)
Drug-related hepatic disorders§	1 (3%)	4 (11%)	1 (3%)	0	1 (3%)	2 (5%)	1 (3%)
Hypoglycaemia§¶	0	0	0	0	1 (3%)	0	0
Allergic reactions§	0	0	1 (3%)	2 (5%)	2 (5%)	0	1 (3%)
Medication errors, abuse, and misuse§	2 (5%)	1 (3%)	1 (3%)	2 (5%)	4 (11%)	2 (5%)	2 (6%)
Pancreatitis§	0	0	0	0	0	0	0
Neoplasms	1 (3%)	3 (8%)	1 (3%)	0	0	2 (5%)	1 (3%)
Malignant tumours§	1 (3%)	0	0	0	0	0	0
Retinal disorders	3 (8%)	1 (3%)	2 (5%)	3 (8%)	3 (8%)	0	1 (3%)
Acute gallstone disease§	0	1 (3%)	1 (3%)	0	0	0	1 (3%)

Data are the number of participants with an event (%). The safety analysis set included all randomly assigned participants. MedDRA=Medical Dictionary for Regulatory Activities. *Not recovered by the data cutoff date of Nov 3, 2025. †In any treatment group. ‡Evaluated using prespecified standardised MedDRA search queries or customised clusters of MedDRA preferred terms. §Values are for adverse events of special interest that occurred during study treatment; all other values are for the trial period (treatment period plus follow-up period). ¶Severe hypoglycaemia (level 3).

Table 3: Participants with adverse events from baseline to week 40 (safety analysis set)

significant differences were observed for the other dose groups (appendix p 11).

The change in eGFR from baseline to week 36 ranged from 1·1 mL/min per 1·73 m² to 10·9 mL/min per 1·73 m² for zenagamtide compared with -1·7 mL/min per 1·73 m² for placebo (table 2). The treatment difference with zenagamtide compared with placebo was statistically significant for all doses except 0·4 mg and 1·5 mg

(table 2). Although the UACR decreased with zenagamtide from baseline to week 36, the change in UACR was only significant for zenagamtide 5 mg and 20 mg compared with placebo (table 2). At week 36, two participants without macroalbuminuria (UACR <300 mg/g) at baseline (week 0) developed new-onset macroalbuminuria (UACR ≥300 mg/g) in each of the zenagamtide 1·5 mg, the zenagamtide 5 mg, and the

placebo groups (six participants in total). Of those participants who had macroalbuminuria at baseline, 31 no longer had macroalbuminuria at week 36 (29 across the zenagamtide groups and two in the placebo group; appendix p 12).

The incidence of adverse events throughout the trial is reported in table 3. Most adverse events were mild to moderate in severity and were recovered from by data cutoff. The most frequently reported adverse events were gastrointestinal in nature (nausea, vomiting, and diarrhoea). Nausea was reported in four (11%) of 38 participants in the zenagamtide 0.4 mg group and 19 (50%) of 38 participants in the 20 mg group, vomiting in one (3%) of 38 participants in the 0.4 mg group and 11 (29%) of 38 participants in the 40 mg group, and diarrhoea in two (5%) of 38 participants in the 0.4 mg group and 14 (37%) of 38 participants in the 40 mg group, representing the lowest and highest observed incidences across the zenagamtide dose groups (table 3). In the placebo group, nausea, vomiting, and diarrhoea were reported in five (14%), two (6%), and two (6%) of 36 participants, respectively. The proportion of participants with gastrointestinal adverse events by severity over time is shown in the appendix (p 19). The time to onset of gastrointestinal adverse events is shown in the appendix (p 20). The time to onset of the first event showed that there was a higher frequency of gastrointestinal adverse events in the 40 mg group than in the other treatment groups from week 0, reaching a plateau at week 30. From week 0 to week 8, higher rates of gastrointestinal adverse events were reported in the 5 mg and 40 mg groups than in the other zenagamtide groups and the placebo group. Serious adverse events were reported in 21 (8%) of 261 participants: four (11%) in the 0.4 mg group, three (8%) in the 1.5 mg group, two (5%) in the 5 mg group, five (13%) in the 10 mg group, three (8%) in the 20 mg group, one (3%) in the 40 mg group, and three (8%) in the placebo group. One severe hypoglycaemic episode (level 3) was reported in one participant in the 20 mg group. No deaths occurred during the trial (table 3).

Discussion

In this dose-finding phase 2 trial, representing the first clinical evaluation of once-weekly subcutaneous zenagamtide in individuals with type 2 diabetes, treatment with zenagamtide 0.4–40 mg resulted in clinically meaningful and statistically significant improvements in HbA_{1c} from baseline to week 36 with all investigated doses compared with placebo, with a significant dose–response relationship. Importantly, a significant decrease in bodyweight compared with placebo was also observed from baseline to week 36 for all zenagamtide doses except 0.4 mg. These results are consistent with the complementary mechanism of action of zenagamtide, a novel unimolecular agonist of GLP-1, amylin, and calcitonin receptors that targets glucose

regulation, weight loss, and potentially broader metabolic health.²¹ The safety and tolerability profile of zenagamtide was overall consistent with the known safety profile of other GLP-1-based and amylin-based therapies.^{5,6}

The significant reductions in HbA_{1c} were observed with zenagamtide across all doses compared with placebo despite the relatively low baseline HbA_{1c} of 7.8% across all groups and the very short duration of exposure to the highest doses of zenagamtide (20 mg for 8 weeks and 40 mg for only 4 weeks). Moreover, these reductions occurred in the context of background SGLT2 inhibitor use in up to 43% of participants across treatment groups, indicating strong glycaemic efficacy. The efficacy of zenagamtide was further supported by the higher proportion of participants attaining clinically relevant HbA_{1c} targets (<7% and ≤6.5%) with all zenagamtide doses compared with placebo.

The reductions in HbA_{1c} reported in the current trial of up to 1.7% after 36 weeks, even after only 4 weeks of receiving the highest dose of 40 mg, are broadly in line with those previously reported for GLP-1 receptor agonists in phase 3 trials. In phase 3 trials for semaglutide and tirzepatide, HbA_{1c} reductions ranging from 2.0% to 2.6% have been reported; however, in these phase 3 trials, there were higher baseline HbA_{1c} levels, fewer users of SGLT2 inhibitors, and longer trial durations than in the current trial.^{5,6,26,27}

Results from phase 3 trials in participants with type 2 diabetes showed reductions in HbA_{1c} of up to 2.3% after 68 weeks of treatment with CagriSema.^{17,18} In individuals with overweight or obesity and type 2 diabetes, a HbA_{1c} reduction of 1.9% after 68 weeks of treatment was reported for CagriSema;²⁸ however, indirect comparisons across trials should be interpreted with caution owing to differences in trial design, duration, baseline characteristics, and background therapy.

Clinically meaningful improvements were seen in the percentage of TIR of up to 30.4% from baseline to week 36 with zenagamtide across all doses compared with placebo, equating to up to 7 h 18 min additional time spent in target range per day. A 3% increase in mean percentage of TIR is considered a clinically meaningful improvement for individuals with type 2 diabetes.²⁹ In addition, the percentage of TIR was higher than the internationally recommended target of above 70% across all zenagamtide treatment groups at week 36.²⁹ Mean percentage of TITR almost doubled by week 36 versus baseline for the majority of zenagamtide doses. In general, the mean percentage of TBR remained low across all zenagamtide doses: at week 36, the mean percentage of TBR below 3.9 mmol/L (70 mg/dL) and TBR below 3.0 mmol/L (54 mg/dL) remained below the internationally recommended target of 4% and 1%, respectively, except in the 10 mg group, in which longer TBR (4.1% and 3.2%, respectively) was driven by one participant. No adverse events or hypoglycaemic episodes were reported for this participant.²⁹

Significantly greater reductions in bodyweight of up to -14.6% were observed in participants receiving zenagamtide compared with those receiving placebo (-2.1%) with no signs of plateau; the magnitude of weight loss observed was broadly consistent with that reported for other incretin-based therapies in similar type 2 diabetes populations.^{17,18} Weight reductions of up to -14.2% after 68 weeks of treatment have been reported with CagriSema in participants with type 2 diabetes.^{17,18} Notably, in the current trial, there was a higher proportion of men than women enrolled (66% vs 34%), which could have influenced the observed weight loss outcomes because men typically have less weight loss than women.³⁰ The robust reductions in bodyweight observed with zenagamtide in this trial support the previous findings for this molecule. In a phase 1b/2a trial of once-weekly subcutaneous zenagamtide in individuals with overweight or obesity without type 2 diabetes, the highest dose of zenagamtide (60 mg) was well tolerated, resulting in a mean percentage change in bodyweight of -24.3% after 36 weeks with no sign of plateau.²⁰ These findings are important because the health benefits derived from weight loss in people living with type 2 diabetes extend beyond improvements in glycaemic control.^{12,31} A 5–10% reduction in bodyweight results in improvements in cardiovascular risk factors, whereas a reduction of 10–15% can lead to improved metabolic health and a reduced risk of long-term complications associated with type 2 diabetes.^{1,31,32}

Estimated zenagamtide-to-placebo treatment ratios for hsCRP generally decreased across the subcutaneous zenagamtide dose range, with lower ratios at higher doses. These data should be interpreted with caution owing to small participant numbers and the wide and overlapping CIs. A significant increase in eGFR of up to 11 mL/min per 1.73 m^2 was observed with 10 mg zenagamtide with no corresponding significant change in UACR. However, the increase in eGFR should be interpreted with caution because eGFR was within the normal range at baseline (mean eGFR $84\text{--}92\text{ mL/min per }1.73\text{ m}^2$). To better assess improvement in kidney function, zenagamtide should be investigated further in a study designed and powered to assess mechanisms for improvement in kidney function. It is also reassuring to see some potential effects on blood pressure and some lipid parameters (eg, triglycerides) with zenagamtide treatment compared with placebo because hypertension and dyslipidaemia are both common comorbidities that have a complex interrelationship with type 2 diabetes and obesity. The increase in pulse rate from baseline to week 36 across zenagamtide dose groups was primarily observed during dose escalation with no clear dose-dependent trend. In the current trial, the observed increase in pulse rate was slightly higher than that seen with tirzepatide and semaglutide in trials of people with type 2 diabetes,^{5,26} but is aligned with findings from the zenagamtide phase 1b/2a study.²⁰

The most common adverse events were gastrointestinal and mild to moderate in severity, as aligned with previous trials for GLP-1 and amylin receptor agonists.^{5,6,26}

Throughout the trial, most treatment discontinuations were reported in the higher dose groups (34–45% in the 10–40 mg groups), many of which were related to nausea and vomiting. Most treatment discontinuations occurred during dose escalation of zenagamtide. This trial used a fixed-dose escalation protocol, whereby if an individual did not tolerate a given dose they were discontinued from treatment, rather than receiving a dose reduction. Because participants were assigned to predefined dose regimens according to the randomisation schedule and were required to escalate without protocol-permitted dose reduction, intolerance at a given dose more frequently resulted in discontinuation, which probably contributed to the higher withdrawal rates observed. In other clinical studies in which dose reduction and re-escalation strategies are permitted, improved treatment persistence and tolerability might be expected.

This trial had several strengths. Evaluation of subcutaneous zenagamtide across a broad dose range enabled characterisation of a possible dose–response relationship to support phase 3 dose selection. Importantly, the trial design allowed assessment of the balance between efficacy, safety, and tolerability (ie, key considerations in early clinical development) while also informing the potential value of this novel therapeutic approach. The double-blind design reduced assessment bias, and use of masked CGM data enabled a comprehensive evaluation of glycaemic control, including clinically meaningful TIR metrics.

Several limitations should also be considered. The 36-week trial duration restricted the assessment of the long-term glycaemic durability, weight trajectory, and safety associated with subcutaneous zenagamtide. Exposure to the highest doses was brief (20 mg for 8 weeks and 40 mg for 4 weeks), which could have constrained a full evaluation of the efficacy and tolerability at these doses, given that HbA_{1c} measurements usually reflect the average blood glucose during the preceding 8–12-week period. Given the observed dose-dependent discontinuation, multiple imputation under a missing-at-random assumption could have preferentially reflected participants who tolerated higher doses; therefore, treatment effects at these doses might not be generalisable to the full randomly assigned population. In addition, treatment group sample sizes were relatively small compared with those used in phase 3 trials. Finally, although participants and investigators were masked to treatment allocation, they were not masked to dose level. This partial masking could have introduced expectation-related effects, which might have influenced adverse event reporting and treatment discontinuation.

In conclusion, in people with type 2 diabetes, once-weekly subcutaneous zenagamtide 1.5 mg to 40 mg

showed clinically meaningful and significant improvements in glycaemic control (HbA_{1c} and TIR) and bodyweight compared with placebo, with positive improvements in other cardiorenal risk factors and a safety and tolerability profile consistent with those of other GLP-1-based and amylin-based therapies. Additional mechanistic studies, complemented by long-term clinical trials, are necessary to quantify the relative contributions of GLP-1, amylin, and calcitonin receptor agonism to the emerging benefits and sustained safety of zenaglutide. Overall, these results support further investigation of once-weekly subcutaneous zenaglutide in phase 3 trials.

Contributors

PM, MB, and JR were trial investigators and helped to obtain data. LDH and A-LFC were responsible for medical oversight during the trial. LNV had access to the raw data and was responsible for the statistical considerations in the analysis and trial design. PM, MA, LNV, and LDH accessed and verified the data. All authors participated in reviewing and interpreting the data outputs and in developing the manuscript. All authors had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Declaration of interests

PM has received honoraria as a consultant for Abbott, Biomea Fusion, Dexcom, Insulet, Minimed (Medtronic), and Novo Nordisk, and is a member of the speakers bureau for Abbott, Dexcom, Insulet, Minimed (Medtronic), and Novo Nordisk. VRA reports institutional contracts with Amgen, Applied Therapeutics, AstraZeneca, Biomea Fusion, Corcept Therapeutics, Eli Lilly, Fractyl Health, Kailera Therapeutics, Novo Nordisk, Pfizer, Recordati, Rhythm Pharmaceuticals, and Servier, and has received consultancy fees from the Baim Institute for Clinical Research, Mediflix, Roche, and Sanofi. MB has received honoraria as a consultant and speaker from Abbott, Amgen, AstraZeneca, Bayer, Boehringer Ingelheim, Daiichi-Sankyo, Lilly, MSD, Novo Nordisk, Novartis, Pfizer, and Sanofi, and reports chairing a clinical trial data safety monitoring board for Boehringer Ingelheim. MA, A-LFC, LNV, and LDH are employees and shareholders of Novo Nordisk. JR reports clinical research grants from Amgen, Applied Therapeutics, AstraZeneca, Biomea Fusion, Boehringer Ingelheim, Corcept Therapeutics, Corxel, Eli Lilly, Hanmi Pharmaceutical, Merck, Novo Nordisk, Pfizer, Regeneron, Roche, Regor Therapeutics, Sanofi, Structure Therapeutics, and Terns Pharma; serves or has served on scientific advisory boards for, and has received honoraria or consulting fees from, Amgen, Applied Therapeutics, Arrowhead Pharmaceuticals, Biomea Fusion, Boehringer Ingelheim, Corcept, Eccogene, Eli Lilly, Hanmi Pharmaceutical, iZO Therapeutics, Novo Nordisk, Regeneron, Roche, Sanofi, Structure Therapeutics, and Terns Pharma; and has received honoraria for lectures from Eli Lilly, Novo Nordisk, and Sanofi.

Data sharing

An authorised researcher can request access to clinical trial data by submitting a research proposal for review and approval by Novo Nordisk and an internal independent review panel. Requests are usually considered after the research is finished and the main results have been published. If the research supports a regulatory application, requests will be considered after the product and its intended use are approved in both the EU and the USA. Participants' clinical data will be anonymised, following an approved internal process, before data are shared to external third parties. Details on how to request access to clinical data are online (novonordisk-trials.com).

Acknowledgments

This trial was funded by Novo Nordisk. Medical writers (Jen Geatrell and Samuel Bestall, Oxford PharmaGenesis, Oxford, UK), funded by Novo Nordisk, assisted with drafting of the manuscript under the direction of the authors. We thank all trial participants, investigators, and trial site staff who were involved in the conduct of this trial.

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